

1. Which compound contains a 1:2 ratio of cation to anion?
 - A) Lithium bromide
 - B) Calcium sulfate
 - C) Magnesium chloride
 - D) Sodium nitrate
 - E) Strontium oxide

2. How many protons, neutrons, and electrons are in one ion of $^{15}\text{O}^{2-}$?
 - A) 15 protons, 13 neutrons, and 8 electrons
 - B) 8 protons, 7 neutrons, and 10 electrons
 - C) 0 protons, 15 neutrons, and 2 electrons
 - D) 15 protons, 0 neutrons, and 13 electrons
 - E) 8 protons, 7 neutrons, and 6 electrons

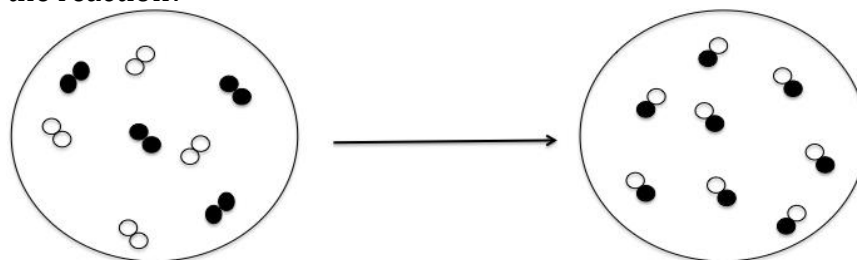
3. In the laboratory, a student recreates the experiments of the photoelectric effect. Which observation correctly correlates to the experiments?
 - A) Electrons are ejected from the metal surface at all frequencies of the incident light.
 - B) The number of electrons ejected increases linearly with the frequency of the incident light, as long as the frequency of the light is above a certain value.
 - C) The kinetic energy of the ejected electrons increases linearly with the intensity of the incident light as long as the frequency of the light is above a certain value.
 - D) The number of electrons ejected increases linearly with the intensity of incident light, as long as the frequency of the light is above a certain value.
 - E) The kinetic energy of ejected electrons is independent of the frequency of incident light.

4. A new element was discovered and found to have atomic mass of 342.38 amu. This element has two isotopes, the lighter of which has a mass of 340.91 amu and an abundance of 68.322%. What is the mass of the heavier isotope in amu?
 - A) 341.65
 - B) 341.12
 - C) 345.55
 - D) 348.31
 - E) 576.30

5. Which of the following is the proper name of a covalent compound?

- A) Iron(III) nitrate
- B) Phosphate ion
- C) Nickel(II) sulfate
- D) Sulfur trioxide
- E) Mononitrogen dioxide

6. The scenes below represent a chemical reaction between elements A (open circles) and B (closed circles). Which best represents the balanced equation for the reaction?



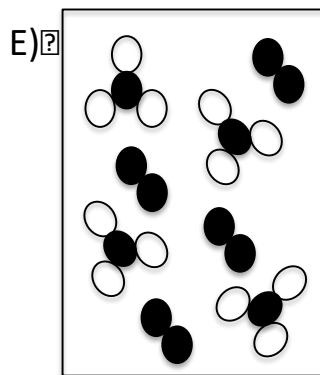
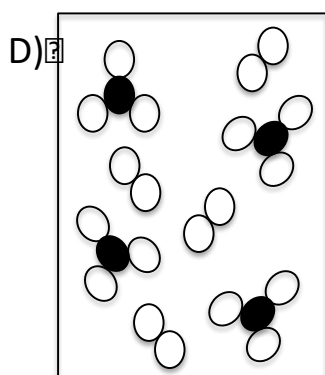
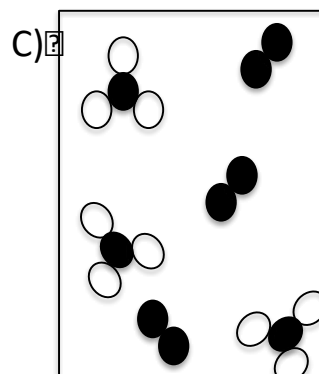
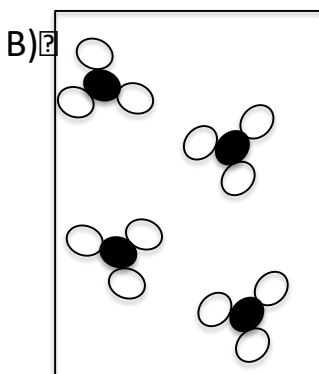
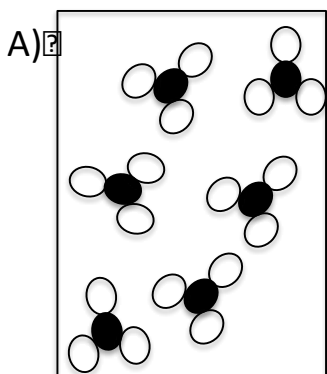
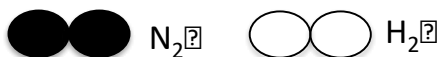
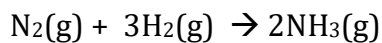
- A) $2A + 2B \rightarrow A_2 + B_2$
- B) $A_2 + B_2 \rightarrow 2AB$
- C) $B_2 + 2AB \rightarrow 2B_2 + A_2$
- D) $4A_2 + 4B_2 \rightarrow 8AB$
- E) $A + B \rightarrow AB$

7. Which statement is true regarding the elements Al, Ag, and Au?

- A) One mole of Ag has greater mass than one mole of Au.
- B) A 200. g sample of Au contains fewer atoms than a 200. g sample of Al.
- C) One mole of Al has the same mass as one mole of Ag.
- D) A 100. g sample of Al contains the same number of atoms as a 100. g sample of Au.
- E) One atom of Al has greater mass than one atom of Au.

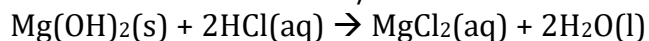
8. Lactic acid forms in muscle tissue during exercise, and it often is blamed (erroneously) for muscle "burn" during intense exercise. Elemental analysis of lactic acid shows that the compound contains (by mass) 40.0% C, 6.71% H, and 53.3% O. If the mass of two moles of lactic acid is 180.16 g, then what is the molecular formula for lactic acid?
- A) CH_2O
B) $\text{C}_3\text{H}_6\text{O}_3$
C) $\text{C}_{3.33}\text{H}_{6.66}\text{O}_{3.33}$
D) $\text{C}_6\text{H}_8\text{O}_8$
E) $\text{C}_4\text{H}_{10}\text{O}_2$
9. The chemical formula for caffeine is $\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$. If you consume 0.75 g of carbon from caffeine, then what mass of caffeine did you consume?
- A) 0.50 g
B) 1.5 g
C) 6.0 g
D) 8.8 g
E) 97 g
10. In the laboratory, a student mixes aqueous solutions of sodium carbonate and calcium nitrate. Which statement is true?
- A) There is no reaction.
B) $\text{NaNO}_3(\text{s})$ forms.
C) $\text{CaCO}_3(\text{s})$ forms.
D) $\text{Na}_2\text{Ca}(\text{s})$ forms.
E) $\text{NO}_3\text{CO}_3(\text{s})$ forms.
11. Which statement is true given the balanced molecular equation?
- $$3\text{Ca}(\text{ClO}_4)_2(\text{aq}) + 2\text{Na}_3\text{PO}_4(\text{aq}) \rightarrow \text{Ca}_3(\text{PO}_4)_2(\text{s}) + 6\text{NaClO}_4(\text{aq})$$
- A) Ca^{2+} and PO_4^{3-} are spectator ions.
B) The total ionic equation is $6\text{Na}^+(\text{aq}) + 6\text{ClO}_4^-(\text{aq}) \rightarrow 6\text{NaClO}_4(\text{aq})$
C) Ca^{2+} and ClO_4^- are spectator ions.
D) No reaction occurs.
E) The net ionic equation is $3\text{Ca}^{2+}(\text{aq}) + 2\text{PO}_4^{3-}(\text{aq}) \rightarrow \text{Ca}_3(\text{PO}_4)_2(\text{s})$

12. A reaction vessel initially contains six molecules of N_2 and six molecules of H_2 . Which image best represents the reaction vessel *after* the chemical reaction occurs?



13. What is the mass of MgCl_2 in the solution resulting when 50.6 g of $\text{Mg}(\text{OH})_2$ react with 45.0 g of HCl per the chemical reaction below? Assume 100% yield for the reaction.

Molar masses: $\text{Mg}(\text{OH})_2 = 58.33 \text{ g/mol}$; $\text{HCl} = 36.46 \text{ g/mol}$; $\text{MgCl}_2 = 95.21 \text{ g/mol}$;
 $\text{H}_2\text{O} = 18.02 \text{ g/mol}$



- A) 23.8 g
- B) 58.8 g
- C) 82.6 g
- D) 118 g
- E) 141 g

14. In the laboratory, a researcher takes 10.0 mL of a 0.247 M aqueous solution of KNO_3 and places it in a flask to which she adds enough water to make a total volume of 100.0 mL. Which statement is true about the resulting solution?
- A) The concentration of the solution did not change.
 - B) The volume of solvent did not change.
 - C) The moles of solute present did not change.
 - D) The new concentration is greater than 0.247 M
 - E) The volume of the solution did not change.
15. Which pieces of information are required to calculate the molarity of a solution of an unknown compound in water?
- I. The molar mass of the compound
 - II. The mass of the compound dissolved in water
 - III. The volume of water in the solution
 - IV. The total volume of the solution
- A) I and II only
 - B) I and III only
 - C) I, II, and III only
 - D) II and IV only
 - E) I, II, and IV only
16. If a gas at 1178 torr in a 3.00 L container is heated from 20. °C to 80. °C at constant volume, what is the new pressure in atm?
- A) 1.2 atm
 - B) 1.9 atm
 - C) 2.4 atm
 - D) 6.2 atm
 - E) 8.0 atm

17. A 1.00 L mixture of helium, neon, and argon has a total pressure of 746 mm Hg at 298 K. If the partial pressure of helium is 341 mm Hg and the partial pressure of neon is 112 mm Hg, then what mass of argon is present in the mixture?

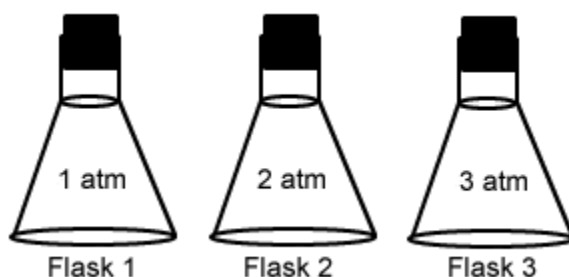
- A) 0.630 g
- B) 4.09 g
- C) 15.7 g
- D) 39.2 g
- E) 479 g

18. Which postulate(s) of the kinetic molecular theory fail(s) for real gases?

- I. The particles of a gas are in constant, random motion.
- II. The volume of each gas particle is negligible in comparison to the volume of the sample.
- III. There are no attractive forces between particles of gas.

- A) I only
- B) II only
- C) III only
- D) II and III only
- E) I and III only

19. Each flask below is the same size and at the same temperature. Which flask contains the most particles of gas?



- A) Flask 1
- B) Flask 2
- C) Flask 3
- D) There is not enough information.
- E) All the flasks contain the same number of particles of gas.

